ANGUS (SHENG-CHUN) CHANG

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Education

Cornell University, Ithaca, NY Expected May 2027
B.S. Mechanical Engineering
Relevant Coursework: Statics & Mechanics of Solids, Mechanics of Materials, Mechanical Design, System Dynamics, Mechatronics, Thermodynamics, Fluid Mechanics

Skills

- **CAD/CAE:** SolidWorks, Fusion 360, Onshape, ANSYS
- **Manufacturing:** FDM 3D printing, CNC machining, tolerancing, fixtures, basic metrology (calipers)
- **Programming:** Python, MATLAB, C++, Arduino, Java
- **UAV:** FPV build + Betaflight tuning; payload integration; rapid iteration workflows
- **Certifications:** FAA Part 107 Remote Pilot; Red Cross Lifeguard-CPR/AED

Projects

UAV Payload Mount & Modular Bracket System 2025
- Designed a modular mounting system in CAD with DFM constraints; reduced part count from **9** to **6** and cut assembly time by **20%**.
- Manufactured and iterated **5** revisions (FDM + CNC); improved fit repeatability and reduced rework by **25%** via tolerance and interface updates.
- Validated through bench vibration checks and flight tests; supported payloads up to **250 g** while maintaining stable tuning.

Test Data Automation Tool (Python/MATLAB) 2024
- Automated parsing and reporting for sensor logs; reduced per-test analysis time and standardized plots/exports for repeatable reporting.

Experience

Autonomous Drone Project Team (CUAD) Sep 2025 – Present
Mechanical Subteam Member
- Designed UAV frame and payload components in CAD; iterated **6** prototypes and reduced structural mass by **38%** while meeting stiffness targets for stable flight.
- Produced drawings with GD&T and maintained a **25+** part BOM; coordinated mounting interfaces with electrical/controls teams to meet envelope and fastener constraints.

Bio-Inspired Fluid Lab, Cornell University Sep 2025 – Present
Undergraduate Researcher, Bat Robot
- Supported CAD modeling and fabrication of lightweight wing structures; reduced assembly mass and improved alignment by via redesigned joints and fixtures.
- Assisted testing across **30+** trials (setup, data capture, summarization); delivered results to guide next iteration.

Cornell University Sep 2024 – Jan 2025
Undergraduate Researcher, Clustering Greenland Seismological Data
- Created a Python pipeline to preprocess **8,000+** seismic events and produce **14** learned/engineered features with automated visualization outputs.
- Performed unsupervised clustering using CNN embeddings + K-means; assessed clustering with silhouette score and stability checks, reducing manual analysis time.

Research & Intellectual Property

- Artificial Leaf device generating oxygen and electricity (patents: Taiwan & Germany, issued)
- Fabric structure based on protein denaturation and adhesion (patents: Taiwan, issued)
- Tunable reflectance using phase-change metamaterials for structural color filtering

Honors & Publications

- Top 300 Regeneron Science Talent Search Scholar (2023)
- ISEF 1st Place Chemistry Special Award (2022)
- Published in *International Journal of High School Research*, Vol. 5, Issue 3